

# Tiered Advertiser Feature Catalog + AI PMF Scoreboard

Built 2026-09-05. Turns the advertiser-facing revenue streams into add-on features across Tiers 1–3, and adds an AI product-market-fit scoreboard that ranks features by retention. Decisions locked with the owner. Values are **advertising value delivered** / **measured activity** — never a revenue or ROI promise. Not legal advice.

## Decisions (locked)

1. **Scope — advertiser features only.** Only advertiser-facing streams become tier features (clean fit for the advertiser tiers). User-facing streams (cosmetics, boosts) and structural fees are **tracked** by the scoreboard but are not tier features.
2. **Pricing — hold prices, ratio climbs.** Adding live features raises each tier's delivered-value stack; the price is held, so the value **ratio** climbs (more value for the same money). No price change, no ROI claim.
3. **PMF weighting — retention-weighted.** The score leans hardest on retention lift (do adopters come back), the truest PMF signal; adoption, engagement, and per-feature revenue fill in the rest.
4. **Founding role — a measured privilege, not a quota.** Founding / pre-revenue Tier 1 gets the **whole catalog free** and is the PMF panel; the AI *observes* what founders use — it never enforces a quota (which keeps it clear of the “paying for feedback” line). **New 2026-09-11:** on top of that free catalog, a founding advertiser ( `is_founding` ) now also resolves to the **Tier 3 / Unlimited capability level for every tier-gated feature** — the most scaled-up version of each, free (e.g. max AI Creative Suite generations, all ad formats, unlimited concurrent experiments, multivariate testing, the full autonomy ceiling, predictive learning, image/video/brand-kit/localization). Gated by `FOUNDING_MAX_SCALE_ENABLED` (default **ON**). This maxes feature **capability**, **not** delivered ad-impression **volume** — the impression allotment is unchanged.

## The tiered feature catalog

`backend/sdk/advertiser-features.ts` — each advertiser-facing revenue stream is an add-on feature with a tier, a conventional value, a readiness (**live** / **gated** / **counsel**), and the `RevenueEvent` `type` it books to. A gated/counsel feature is listed as “*included — activates when its prerequisite lands*” and contributes **\$0** of claimed value until it is live (same rule as the value stacks: every line respects its own toggle).

- **Shared across all tiers:** PPC network advertising on the premium **AdGrid** (catalog key `ppc_grid_placement`, tier 1 — which in this catalog means *included in Tiers 1, 2 AND 3* — status **live**, category **advertising**). Every advertiser, Tier 1 (Founding), Tier 2 (Scale) and Tier 3 (Unlimited), can run their product on the AdGrid: users watch it, answer its questions, and are credited, and the advertiser pays per engagement (PPC). (*Added 2026-09-11.*)
- **Tier 1:** sponsored survey campaign, audience pulse, in-app interstitials, lead-gen program.
- **Tier 2 “Scale”:** Pro audience panels, brand-lift studies, competitive reports, plus gated offerwall/CPA, rewarded video, sponsored push/email, API access, self-serve AI creative studio.
- **Tier 3 “Unlimited”:** product-testing panel, white-label/RaaS, survey-routing arbitrage — all tiers below.
- **Founding:** the entire catalog, free — **plus** every tier-gated feature dialed to its Tier 3 / Unlimited capability ceiling ( `FOUNDING_MAX_SCALE_ENABLED`, default on; caps capability, not impression volume).

`advertiserFeatureCatalog` (read) returns the catalog + a per-tier rollup: live vs. pending counts and the **delivered value added** (what makes the ratio climb, price held). Layered **additively** — it does not mutate the tested Tier 1/2/3 value stacks.

## The AI PMF scoreboard

`backend/sdk/feature-pmf.ts` ranks every feature by a retention-weighted **PMF score (0-100)**:

- **Retention lift** (weight 0.45) — of a feature's prior-window adopters, the fraction active again in the recent window, minus the site baseline, **shrunk toward 0 for low samples** (  $n/(n+K)$  ) so a feature only a handful tried can't top the board on noise.
- **Adoption** (0.20, distinct users), **engagement** (0.15, uses per adopter), **revenue** (0.20, the feature's slice of the one `RevenueEvent` ledger). All admin-tunable ( `PMF_WEIGHT_*`, `PMF_WINDOW_DAYS`, `PMF_SHRINK_K` ).
- **Per-tier revenue ranking** — “which features earn the most, for each tier,” plus the tier's revenue total.
- **Founding segmentation** — usage is logged with a `founding` flag so the panel is measured separately.

**Runs continuously.** `featurePmfScoreboardRun` is on the scheduler every 6h (authorized by the scheduler's signed token, same pattern as the scaling governor), so PMF discovery keeps going after launch — the founding rankings and the operational rankings become a before/after.

## Components

- SDK: `advertiser-features.ts`, `feature-pmf.ts`.
- Functions: `advertiserFeatureCatalog`, `featurePmfScoreboard` (admin read), `featurePmfScoreboardRun` (scheduled build), `featureUsageTrack` (records a use; reads the user's founding flag).
- Entities: `FeatureUsageEvent`, `FeaturePmfSnapshot`.
- Admin page: `src/pages/FeaturePMF.jsx` (ranked scoreboard + per-tier revenue view).
- Schedule: `feature-pmf-scoreboard` (0 \*/6 \* \* \*).
- Settings (Scale & Platform): `ADVERTISER_FEATURE_CATALOG_ENABLED`, `ADVERTISER_FEATURE_CATALOG_JSON`, `FEATURE_PMF_ENABLED`, `PMF_WEIGHT_RETENTION/ADOPTION/ENGAGEMENT/REVENUE`, `PMF_WINDOW_DAYS`, `PMF_SHRINK_K`.

## Complete revenue-stream coverage — all 45 streams (2026-09-05)

Decision (locked): **track all 45 revenue sub-points; tier the advertiser/business-facing ones.** Two layers:

- **Tiered advertiser features (19)** — the advertiser/business-facing streams are add-on features across Tiers 1–3 and are **PMF-ranked** (retention-weighted). This pass added the remaining business-facing ones to the catalog: **B2B SaaS suite** (T2), **managed hosting** and **fraud-as-a-Service** (T3), and **sponsored jackpots** (T1).
- **Full coverage map (all 45)** — `revenue-coverage.ts` + `revenueStreamCoverage` cover **every** sub-point across all 8 categories with its real revenue (its `RevenueEvent` slice), status (built/gated/counsel), live/pending, and whether it's a tiered feature. The user-facing (cosmetics, boosts, season pass), seller-side (commissions, sourcing, affiliate), and structural-fee (BNPL, shipping, FX) streams are **not** advertiser tier features — an advertiser can't "buy" a shipping spread — but they are all **revenue-tracked** so none is invisible. Retention-PMF ranking applies to the advertiser subset (where "do adopters come back" is meaningful); revenue coverage applies to all 45.

Shown on the `FeaturePMF` admin page as the "All revenue streams — coverage" section (grouped by category, with per-category revenue), alongside the retention-weighted scoreboard, the per-tier revenue ranking, and the AI agent plan.

## Usage tracking — auto-wired (2026-09-05)

Adoption/engagement now populate **automatically**. Rather than editing every feature function, the tracker is hooked into `recordRevenue()` (`revenue.ts`) — the shared entry point every advertiser feature already calls to book its revenue. `recordFeatureUseForRevenue()` maps the event's `type` to the catalog feature(s) and logs a `FeatureUsageEvent` (attributing the advertiser). If a caller tags the event with `meta.feature_key` it's precise; otherwise every feature that books that revenue type is credited. Best-effort — it never blocks the ledger. A front-end `featureUsageTrack({ feature_key })` call is still available for surfaces that don't book revenue on use, but the core adoption/engagement now flows on its own.

## The AI PMF & revenue agent (2026-09-05)

`backend/sdk/pmf-agent.ts` + `pmfRevenueAgentRun` (scheduled every 6h, 20 min after the scoreboard) close the loop the owner asked for: an agent that **collects all the feature/site signals, ranks the portfolio for product-market fit and increased revenue, and learns** — while **keeping every existing constraint**.

- **Collect** — writes `pmf_score:<key>` and `feature_revenue:<key>` into the shared `OptimizationSignal` trend store, so the whole AI layer sees them.
- **Plan** — `decideAction()` turns each feature's PMF + revenue into an action: **promote / hold / watch / fix / sunset**, with an advisory pricing hint (raise / hold / lower). Strong fit + above-median revenue → promote and *consider* a price move; strong fit + low revenue → promote to grow adoption first; weak fit + negative retention → sunset candidate; too few adopters → watch (gather more signal).
- **Learn** — durable `AgentLearningMemory` lessons under agent `pmf_revenue_agent`, so it trends in the existing learning dashboards and compounds with the platform's other self-learning.
- **Constraints preserved** — the agent writes signals, learning, and an **advisory plan** to `PmfAgentPlan`; it does **not** auto-change money, pricing, tiers, identity, or legal settings. Every sensitive move (price, sunset, fix-with-discount) is flagged `needs approval` — the same human gate the optimizer and Autonomy Kernel already enforce. No ROI claims; the closed loop is untouched. Runs continuously so discovery keeps improving after launch.

Settings: `PMF_AGENT_ENABLED`, `PMF_AGENT_RECOMMEND_PRICING`, `PMF_AGENT_STRONG_SCORE`, `PMF_AGENT_WEAK_SCORE`, `PMF_AGENT_MIN_SAMPLE`. The plan is shown on the `FeaturePMF` admin page and returned by `featurePmfScoreboard`.

## Value-match to price (every tier delivers ≥ what they pay)

`tierFeatureRollup` now enforces a **value-match-to-price floor** (`ADVERTISER_VALUE_MATCH_TO_PRICE`, on by default): when a tier's summed feature value is below the price the advertiser pays, a guaranteed bonus-inventory top-up (`value_match_to_price_usd`) raises the reported total (`total_offer_value_usd`) to the price. So Tier 1 (~\$27,400 of value vs ~\$13,000 price) is already above and untouched, while Tier 2 tops up to ~\$216,667 and Tier 3 to \$400,000 — each tier's total included value is at least its price. No single feature's conventional value is inflated; the gap is filled only by guaranteed delivery, mirroring the Tier 2 value-match block. The advertiser value-stack table (`FeaturePMF`) shows the top-

up as a distinct “Value-match guarantee” line plus a “Total included value  $\geq$  price ✓” row. Advertising value delivered — never a revenue or ROI promise. See `LIVESTREAM-ADVERTISING-COUNSEL-BRIEF.md` §6.

## Ad-network metrics + demographic & audience targeting (2026-09 update)

**Full network-standard metric set (measured, never guaranteed).** Advertisers now get the complete mobile-ad-network metric set, computed from real platform activity: **eCPM, CPM, IPM, CPP**, and the windowed **ROAS curve (D1 / D3 / D7 / D14 / D28 / D90 / D365)**, plus the publisher-side yield metrics — **fill rate, ARPDAU and retention (D1 / D7 / D28)**. Every figure is measured and shown with its basis; the platform never guarantees an ROI. (Engine: `ad-metrics.ts`; on-demand read: `adMetricsReport`.)

**Tracked over time and AI-optimized.** A scheduled sweep ( `adMetricsSweep` ) tracks each metric as history ( `OptimizationSignal` + `AgentLearningMemory` — no new tables) and runs an AI optimizer that biases delivery from measured results. Every automatic action is reversible and routed through the autonomy kernel ( `ad_optimization` ): it auto-applies on trust or queues a human review, and it **never raises spend and never touches billing or payouts** (those stay permanently gated).

**Demographic + new-vs-existing targeting.** On top of the interest/behavior cohorts, advertisers can target **demographics — age range, gender, country and region** — and **audience type (new vs existing users)**, matched against users’ own non-identifying profile data (never an individual). Advertisers set an `optimize_for` objective — **ROAS, new-user acquisition, or existing-user retention** — that steers the AI delivery optimizer. (Engine: `ad-audience.ts`.)